

Physics Class 11th Solved Model Paper

2026 Smart Syllabus Punjab Boards

1. The ratio of the dimensions of force and energy is:

- (A) T (B) T^{-1} (C) L

$$\frac{F}{W} = \frac{F}{F \cdot d} = \frac{1}{d} \rightarrow L^{-1}$$

2. The projectile gains its maximum height at an angle of:

- (A) 0° (B) 45° (C) 60° (D) 90°

$$h = \frac{v_i^2 \sin^2 \theta}{2g}$$

$\sin 90^\circ = 1$

3. An object in uniform circular motion makes 10 revolutions in 2 seconds. Which of the following statement is true?

- (A) Its period is 2.0 s

$$f = \frac{10 \text{ rev}}{2 \text{ sec}} = 5 \text{ Hz}$$

- (B) Its period is 20 s

$$T = \frac{1}{f} = \frac{1}{5} \text{ sec} = 0.2 \text{ sec}$$

- (C) Its frequency is 5 Hz

- (D) Its frequency is 0.2 Hz

4. If an athlete uses 500 J of energy to lift a load in 2 s, his muscular power is:

- (A) 125 watt

- (B) 250 watt

- (C) 500 watt

- (D) 1000 watt

$$P = \frac{W}{T}$$

$$\frac{500}{2} = 250$$

5. A solid float in water with half of its volume submerged. The density of the solid is:

- (A) equal to the density of water

- (B) half the density of water

- (C) twice the density of water

- (D) zero

$$\rho = \frac{m}{V}$$

Wage = Buoyant force

$$\rho_{\text{solid}} V_{\text{sub}} = \rho_{\text{water}} V_{\text{sub}}$$

6. The pressure will be low when speed of a fluid is:

- (A) zero

- (B) high

- (C) low

- (D) constant

7. First law of thermodynamics is based upon law of conservation of

- (A) mass

- (B) energy

- (C) momentum

- (D) charge

$$\rho \times V = \rho \times \frac{1}{\rho} = 1$$

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{const}$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

$$v_2 > v_1$$

8. The bending of waves around an obstacle is called as:

- (A) refraction

- (B) reflection

- (C) diffraction

- (D) interference



(A) refraction

(B) reflection

(C) diffraction

(D) interference

9. If the amplitude of the wave is tripled, then the amount of energy is increased by:

(A) 3 times

(B) 6 times

(C) 9 times

(D) 12 times

$$E \propto A^2$$

$$E = \frac{1}{2} \rho A^2 v$$

$$A' = 3A$$
$$E' = (3A)^2 = 9A^2$$

10. The mathematical representation of Malus's law is:

(A) $I = I_0 \cos^2 \theta$

(B) $I = I_0 \sin^2 \theta$

(C) $I = I_0 \tan^2 \theta$

(D) $I = I_0 \cot^2 \theta$

$$W = qV = eV = 100 \text{ eV}$$

11. What is the work done on an electron by potential difference of 100 volts?

(A) $1.6 \times 10^{-19} \text{ eV}$

(B) $1.6 \times 10^{-17} \text{ eV}$

(C) $6.25 \times 10^{-17} \text{ eV}$

(D) 100 eV

12. When a wire of resistance R is cut into two equal parts, its resistance becomes $R/2$. What happens to resistivity?

(A) It becomes double

(B) It remains the same

(C) It becomes half

(D) It becomes one fourth

$$1.6 \times 10^{-19} \text{ C}$$
$$\times 10$$
$$1.6 \times 10^{-17}$$

$$R = \frac{\rho L}{A}$$

B → strength of mag. field

13. The unit $\text{NA}^{-1} \text{m}^{-1}$ is called:

(A) weber

(B) tesla

(C) coulomb

(D) none of these

$$F = BIL \Rightarrow B = \frac{F}{I \cdot L} = \frac{N}{A \cdot m} \text{ mag. flux density}$$

14. According to Faraday's law, what causes an emf (voltage) to be generated in a coil?

(A) A constant magnetic field

(B) A changing magnetic flux linkage

(C) Heat applied to the coil

(D) Increasing the wire thickness

$$\mathcal{E} = -N \frac{\Delta \Phi_B}{\Delta t}$$

15. If the rest mass m_0 of a particle increases to m due to its high speed, then its kinetic energy is:

15. If the rest mass m_0 of a particle increases to m due to its high speed, then its kinetic energy is:

(A) $\frac{1}{2} mc^2$

(B) $\frac{1}{2} mv^2$

(C) $(m - m_0)c^2$

(D) $\frac{1}{2} (m - m_0)c^2$

$$E = mc^2 = (m - m_0)c^2$$

16. Baryon is formed by combination of:

(A) 2 quarks

(B) 3 quarks

(C) 4 quarks

(D) a quark and an anti-quark

17. Which one of the following forces has negligible effect between the elementary particles?

(A) Strong nuclear force

(B) Weak nuclear force

(C) Gravitational force

(D) Electromagnetic force

SUBJECTIVE TYPE (Part-1)

Max. Marks: 68

Time: 2:40 hours

Q. 2 Write short answers of any Eight (08) of the following questions: $(2 \times 8 = 16)$

(i) How many significant figures are there in the following measurements?

(a) 37 km

(b) 0.002953 m

(c) 7.50034 cm

(d) 200.0 m

(a) 2

(b) 4

(c) 6

(d) 4

(ii) What is meant by a dimensionless quantity? Give one example.

A quantity having no units and no dimension is called dimensionless quantity.

Example: Refractive index, strain, coefficient of friction.

$$n = \frac{c}{v} \quad \mu = \frac{F_s}{F_n} \quad \pi$$

(iii) Differentiate between an elastic and an inelastic collision.

Elastic collision:

Both momentum and kinetic energy are conserved.

Inelastic collision:

Momentum is conserved but kinetic energy is not conserved.

- (iv) Show that rate of change in momentum is equal to force applied. Also state Newton's second law of motion in terms of momentum.

For a body of mass m with initial velocity v_i and final velocity v_f over a time t , the acceleration a is:

$$a = \frac{v_f - v_i}{t}$$

The acceleration produced by an external force F is:

$$a = \frac{F}{m}$$

Equating the two expressions for acceleration:

$$\frac{F}{m} = \frac{v_f - v_i}{\Delta t}$$

$$F = \frac{mv_f - mv_i}{\Delta t}$$

$$F = \frac{\text{Change in momentum}}{\text{Time}}$$

$$F = \frac{\Delta p}{\Delta t}$$

Statement:

rate of change in momentum of a body is equal to the applied force.

- (v) Mass is a measure of inertia in linear motion. What is its analogue in rotational motion? Describe briefly.

The analogue of mass in rotational motion is Moment of Inertia.

The moment of inertia, denoted by I , is a measure of an object's resistance to a change in its rotational state of motion.

$$I = \sum m_i r_i^2$$



- (vi) Prove the relation; $1 \text{ rad} = 57.3^\circ$

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sq

$$2\pi \text{ rad} = 360^\circ$$

$$1 \text{ rad} = \frac{360^\circ}{2\pi} = \frac{360^\circ}{2 \times 3.1416}$$

$$1 \text{ rad} = 57.3^\circ$$

(vii) How does an astronaut feel weightlessness while orbiting from the Earth in a spaceship?

Astronaut and spaceship are both in free fall around Earth.
Since no normal reaction acts on astronaut, hence apparent weight becomes zero, so astronaut feels weightless.

$T = 0$ ↓

Apparent weight ↑ $N = mg - ma$

$F_{\text{net}} = W - N$

$N = W - F_{\text{net}}$

$N = mg - ma$

$N = mg - mg = 0$

↑ N

$F = ma$
 $F_{\text{net}} = ma$
 $a = g$

" $a = g$ "
 $N = 0$

(viii) When a moving car turns around a corner to the left, in what direction do the occupants tend to fall? Explain briefly.

The occupants tend to fall towards the right due to inertia.

Reason:

Before the turn, the occupants are moving in a straight line. When the car turns left, a centripetal force acts on the car to change its direction. However, according to Newton's first law, the occupants' bodies, by inertia, tend to continue moving in a straight line. This tendency to maintain the original straight path makes them appear to be thrown outward, to the right, relative to the turning car.

(ix) What is a conservative force? Give examples.

A conservative force is a force for which the work done in moving an object between two points is independent of the path taken between the points.

Examples:

- Gravitational force
- Electrostatic force

(x) Calculate power of a crane in kilowatt which lifts a mass of 1000 kg to a height of 100 m in 20 seconds.

Given: $m = 1000 \text{ kg}$
 $h = 100 \text{ m}$
 $t = 20 \text{ sec}$
 $g = 9.8 \text{ ms}^{-2}$

To find Power in KW?

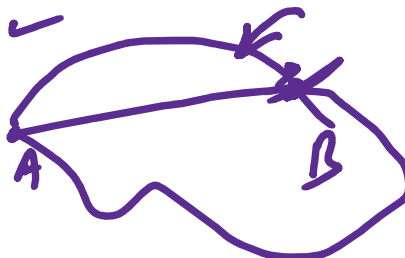
$$P = \frac{W}{t} = \frac{mgh}{t} = \frac{1000 \text{ kg} \times 9.8 \text{ ms}^{-2} \times 100 \text{ m}}{20 \text{ sec}}$$
$$P = 49 \text{ KW}$$

$49000 \text{ W} \times \frac{1 \text{ KW}}{1000}$

(xi) Why work done against friction is non-conservative in nature? Explain briefly.

Because it depends on the path followed, not just on the initial and final positions. In friction, the force always opposes motion. If an object moves between the same two points along different paths, the distance traveled changes, so the work done against friction also changes. Therefore, the work done is path-dependent. Moreover, the energy spent against friction is converted into heat and sound and cannot be fully recovered as mechanical energy. Hence, friction does not conserve mechanical energy.

$$W = F \cdot d$$



(xii) Differentiate between P.E and absolute P.E.

$$P.E = mgh$$

Potential Energy (P.E) can have different values depending on where we

Potential Energy (P.E) can have different values depending on where we choose the zero level.

Absolute Potential Energy always takes zero at infinity, so it gives a true measure of energy stored due to position.

P.E is relative, while absolute P.E is fixed and reference-independent.

$$U = -\frac{GMm}{r}$$

✓ ↑



Q. 3 Write short answers of any Eight (08) of the following questions: (2 × 8 = 16)

- (i) Explain the working of a carburetor of a motorcar using Bernoulli's principle.

The Bernoulli's principle, which states that when the speed of a fluid increases, its pressure decreases.

In the carburetor, Air speeds up in a narrow section (venturi), which causes the pressure to drop (low pressure). This low pressure sucks the fuel into the air stream to mix it correctly.

- (ii) Differentiate between laminar flow and turbulent flow.

Laminar flow

- The particles follow smooth, regular, predictable and parallel paths.
- The velocity of the fluid at any point remains constant.
- Relatively low energy loss due to internal friction.

Turbulent Flow

- The irregular or unsteady flow of the fluid is called turbulent flow.
- The velocity of the fluid at any point changes abruptly with time.
- High energy loss due to the formation of eddies.

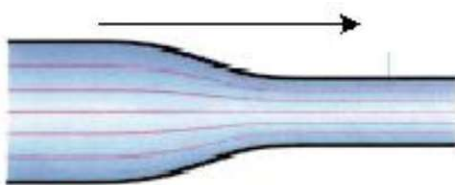


Fig. 5.11: Streamlines (laminar flow)

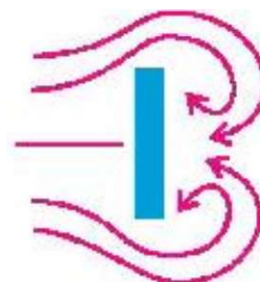


Fig. 5.13: Turbulent flow

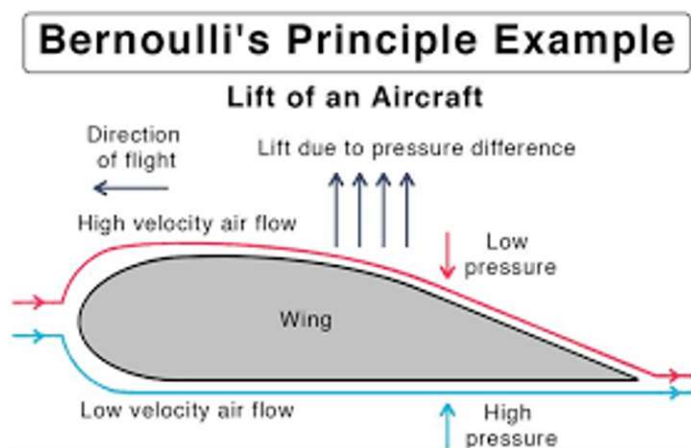
(iii) Why must speed of an object be minimum against drag force?

- As $F_{drag} = 6\pi\eta rv$
- Drag force increases with speed.
- At higher speeds, F_{drag} will not vary linearly due to which more energy losses occur.
- So minimum speed is required to reduce drag force.

(iv) Why wings of an aero plane are rounded outward while flattened inward?

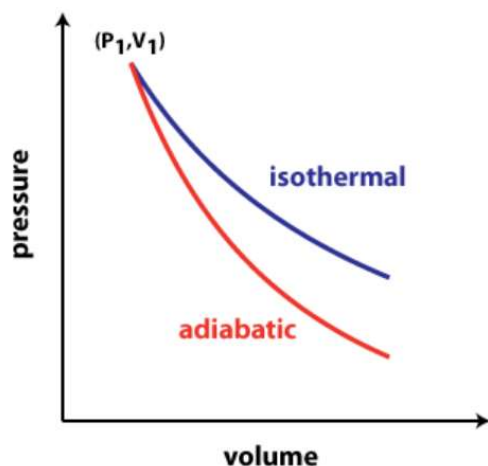
The wings of an aero plane are designed to create a difference in air speed above and below the wing, generating lift due to Bernoulli's principle.

- **Rounded Outward (Upper Surface):** The top surface is more curved (longer path). This forces the air flowing over it to move faster than the air underneath, which lowers the pressure on the top surface.
- **Flattened Inward (Lower Surface):** The bottom surface is flatter (shorter path). This allows the air flowing underneath to move slower than the air on top, resulting in a higher pressure on the bottom surface.



(v) Why adiabat is steeper than isotherm? Explain briefly.

- **Isothermal Process:** Temperature remains constant. When the gas expands, it needs energy to do work. To keep its temperature constant, the gas takes heat from the surroundings. This extra heat helps the gas maintain its energy, so the pressure does not decrease very quickly as volume increases.
- **Adiabatic Process:** No heat exchange takes place. When the gas expands, it does work using its own internal energy, so its temperature decreases, causing pressure to fall more sharply.



- (vi) Explain how thermodynamics relates to the concept of energy conservation.

This is due to 1st Law of Thermodynamics which states that:

In any thermodynamic process, when heat Q is added to a system, this energy appears as an increase in the internal energy ΔU stored in the system plus the work W done by the system on its surroundings.

$$Q = \Delta U + W$$

In simpler terms, energy can neither be created nor destroyed; it can only be transformed from one form (e.g., heat) into another (e.g., internal energy or mechanical work). Thus the total energy in an isolated system remains constant over time.

- (vii) Analyze the relationship between intensity of a wave, area and amplitude.

$$I = \frac{P}{A} \Rightarrow \boxed{I \propto \frac{1}{A}} \rightarrow \text{intensity proportional to area.}$$

$$\boxed{I \propto A^2}$$

I is directly proportional to the square of amplitude.

- (viii) How should a sound source move with respect to an observer so that the frequency of its sound does not change?

Case 1: Relative speed is zero

If there is no relative motion between the sound source and the observer (both are at rest or move together with the same speed in the same direction), the distance

between them does not change. Therefore, no Doppler effect occurs, and the frequency heard by the observer remains the same as the original frequency.

Case 2: Source moves perpendicular to the line joining the observer

If the sound source moves at right angles to the line joining the source and the observer, there is no component of velocity toward or away from the observer. Since only the velocity component along the line of sight causes a Doppler shift, the observed frequency does not change in this case either.

(ix) What conditions must be satisfied in order that two-source interference fringes may be observed?

- Same frequency ✓
- Constant phase difference ✓
- Coherent sources ✓

(x) Can you apply Doppler effect to light waves? Explain briefly.

- Yes, the Doppler effect can be applied to light waves.
- Explanation: The Doppler effect for light waves (electromagnetic waves) is the change in the observed frequency/wavelength of light due to the relative motion between the source of light and the observer.
 - If the source is moving towards the observer, the observed wavelength decreases (frequency increases), which is called blueshift.
 - If the source is moving away from the observer, the observed wavelength increases (frequency decreases), which is called redshift. This is a crucial tool in astronomy for determining the motion of stars and galaxies.

(xi) Compare Brewster's law and Malus's law.

Brewster's Law

- It states that when unpolarized light is incident on a transparent surface at a certain angle (called Brewster angle), the reflected light becomes completely plane-polarized.
- Related to Reflection.
- Determines the angle of incidence θ_B that is $\tan(\theta_B) = n$

Malus's Law

- It states that the intensity of plane-polarized light transmitted through an analyzer depends on the angle between the light's polarization direction and the analyzer's axis.
- Related to transmission.
- Determines the intensity of plane polarized light passing through an analyzer that is $I = I_0 \cos^2(\theta)$

(xii) How do the properties of a system, such as masses, orbital period, and eccentricity, affect the characteristics of gravitational waves?

- The amplitude of the gravitational waves is directly proportional to the total mass of the system that is $A_{GW} \propto m$
- The frequency of the gravitational waves is inversely proportional to the orbital period of gravitational waves. $f_{GW} \propto \frac{1}{T}$
- Eccentricity determines the shape and complexity of the GWs. A low-eccentricity (nearly circular) orbit produces a nearly sinusoidal wave, while a highly eccentric orbit produces a more complex, non-sinusoidal waveform.

Q. 4 Write short answers of any Six (06) of the following questions: (2 × 6 = 12)

(i) Is electron-volt a unit of potential difference or energy? Explain.

Electron volt is the unit of energy.

→ **The electron-volt is defined as the amount of kinetic energy gained by a single electron when it is accelerated through an electric potential difference of one volt. $W = qV$**
 $1eV = 1.602 \times 10^{-19} J$

$$W = qV$$

$$\frac{W}{q} = V$$

$$V = \frac{W}{q} = \frac{eV}{e} = V$$

(ii) What is conventional current? Explain briefly.

- Conventional current is the hypothetical flow of positive charge.
- Conventional current flows from the positive terminal to the negative terminal of a source.
- Its direction is opposite to the flow of electronic current.

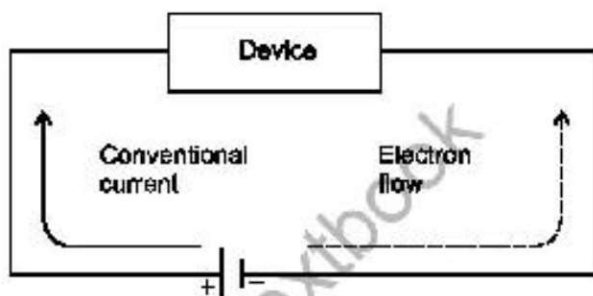
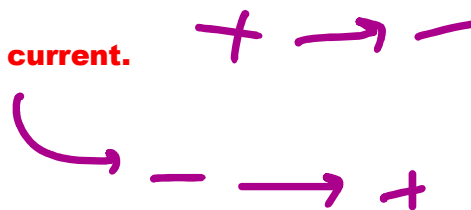


Fig. 9.21

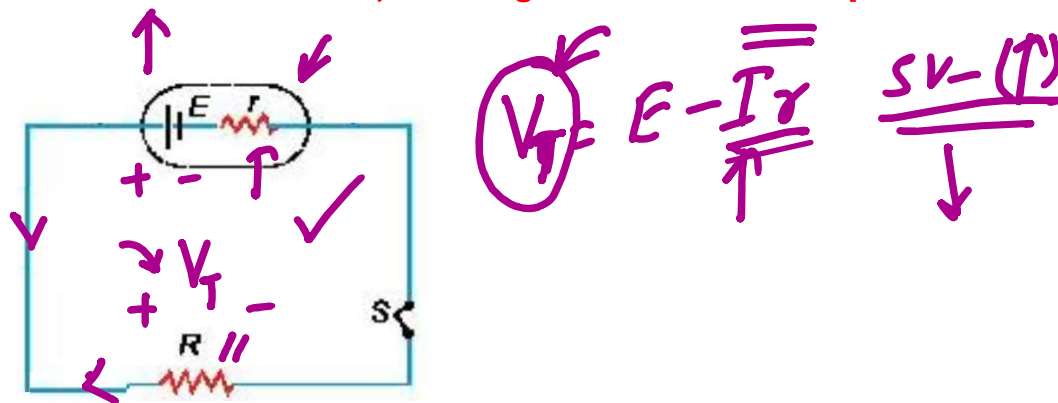


(iii) Why does the terminal potential difference of a battery decrease when the current drawn from it is increased?

The terminal potential difference is given by the equation:

$V_T = E - Ir$, where **E** is the emf of the battery, V_T is the terminal P.D, Ir represents the voltage drop across the internal resistance

From above equation, as the current I drawn increases, the internal voltage drop (Ir) increases, which subtracts from the EMF, resulting in a lower terminal potential difference (V_T).



(iv) Does the induced emf always act to decrease the magnetic flux through a circuit?

- No, the induced emf does not always act to decrease the magnetic flux.

→ According to Lenz's law, the induced emf always acts in such a direction that it opposes the change in magnetic flux.

- If the magnetic flux is increasing, the induced emf produces a current that opposes the increase that is tries to reduce magnetic.
- If the magnetic flux is decreasing, the induced emf produces a current that opposes the decrease that is tries to increase the magnetic flux.

(v) Distinguish between magnetic flux and flux density.

Magnetic Flux (ϕ_B)

- The total number of magnetic field lines passing normally through a given surface area (A)
- $\phi_B = B \cdot A = BA \cos \theta$
- A scalar quantity.
- Unit is Weber (Wb)

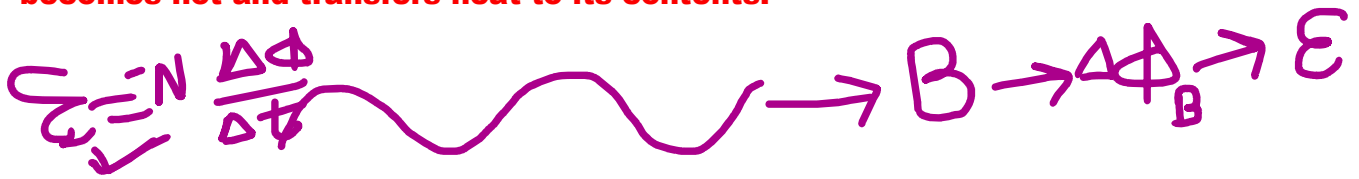
Magnetic Flux Density (B)

- The measure of the strength of the magnetic field at a specific point also called Magnetic field strength.
- $B = \frac{\phi_B}{A}$, when B is uniform and Perpendicular to A
- A vector quantity.
- Unit is Telse ($T = \frac{Wb}{m^2}$)

(vi) How does an electric heater work on the principle of electromagnetic induction?

When an alternating current (AC) flows through a coil inside the heater, it produces a continuously changing magnetic field around it. If a conducting object (such as a metal vessel) is placed in this changing magnetic field, the magnetic flux linked with the object changes. According to Faraday's law of electromagnetic induction, an emf is induced in the conductor.

This induced emf causes eddy currents (currents moving in loops inside the conductor) to flow in the metal. Due to the electrical resistance of the metal, these currents produce heat (Joule heating $\rightarrow H = I^2 R t$). As a result, the metal vessel becomes hot and transfers heat to its contents.



Electron

- Exists outside the nucleus in atoms.
- Present in the electron cloud of an atom.
- Usually much lower (depends on energy).

(ix) Why neutrino must be released in the positron emission?

Positron emission (β^+ decay)

- In positron emission, a proton inside a nucleus transforms into a neutron, emitting a positron (e^+):

$p \rightarrow n + e^+ + \bar{\nu}_e$ here, $\bar{\nu}_e$ is an electron neutrino.

- A neutrino must be emitted in positron emission to simultaneously conserve:

1. Energy
2. Momentum ✓
3. Angular momentum (spin)
4. Lepton number

Without the neutrino, these fundamental conservation laws would be violated.

Note: Attempt any Three (03) questions. Each question carries eight (08) marks.

Q. No 5 (a) Give the dimensions of velocity. Explain elastic collision in one dimension and derive the final velocities. (5)

(b) A car is moving with a speed of 108 km h^{-1} . If its wheel has a diameter of 60 cm, find its angular speed in rad s^{-1} and rev s^{-1} . (3)

$V = [L T^{-1}]$ $v_{\text{rev}} = \frac{\text{displ}}{T} = \frac{[L]}{[T]}$

b) Given data

$v = 108 \text{ km h}^{-1}$

diameter = $60 \text{ cm} = 0.60 \text{ m}$

To find
Angular speed (ω) in rad s^{-1} & rev s^{-1}

Sol. As we know that

$v = r\omega$

$\omega = \frac{v}{r} \rightarrow \textcircled{1}$

$v = \frac{108 \text{ km}}{\text{hr}} = \frac{108 \times 10^3 \text{ m}}{3600 \text{ sec}}$
 $\text{h) } - 30 \text{ m s}^{-1} \checkmark$

$$\boxed{v = 30 \text{ m s}^{-1}} \checkmark$$

$$d = 2r \Rightarrow r = \frac{d}{2} = \frac{0.6 \text{ m}}{2}$$

$$\Rightarrow \boxed{r = 0.30 \text{ m}} \checkmark$$

$$\textcircled{1} \Rightarrow \omega = \frac{30 \text{ m s}^{-1}}{0.30 \text{ m}}$$

$$\boxed{\omega = 100 \text{ rad s}^{-1}}$$



Angular speed in rev s⁻¹

$$1 \text{ rev} = 2\pi \text{ rad}$$

$$\boxed{1 \text{ rad} = \frac{1}{2\pi} \text{ rev}}$$

$$\omega = \frac{100 \text{ rev s}^{-1}}{2\pi}$$

$$\boxed{\omega = 16 \text{ rev s}^{-1}}$$

Q. No 6 (a) State and explain the work–energy theorem. How is the mechanical work expressed in terms of pressure and volume for a gas? Derive the equation. (5)

(b) A spring with a spring constant 200 N m⁻¹ is stretched by 0.5 m. Find the elastic *P.E.* stored in the spring. (3)

(b) A spring with a spring constant 200 N m^{-1} is stretched by 0.5 m . Find the elastic $P.E.$ stored in the spring. (3)

Given $K = 200 \text{ N m}^{-1}$
 $x = 0.5 \text{ m}$
To find! $U = ?$
Sol

$$U = \frac{1}{2} K x^2$$

$$= \frac{1}{2} (200 \text{ N m}^{-1}) (0.5 \text{ m})^2$$

$U = 25 \text{ J}$

Q. No 7 (a). What are gravitational waves? How they can be predicted and detected? Also, give their four basic types. (5)

(b) Two trucks P and Q are travelling along a motorway in the same direction. The leading truck 'P' travels at a steady speed of 12 ms^{-1} , the other truck Q, travelling at a steady speed of 20 ms^{-1} , sounds its horn to emit a steady note which P's driver estimate, has a frequency of 830 Hz . What frequency does Q's own driver hear? (Speed of sound = 340 ms^{-1}). (3)

Q. No 8 (a) What are ferrofluids. Give their working, uses and applications in detail. (5)

(b) A particle of charge $+20 \mu\text{C}$ is placed between two parallel plates, 10 cm apart and having a potential difference of 0.5 kV between them. Calculate the electric field between the plates, and the electric force exerted on the charged particle. (3)

Given data ✓ $q = 20 \mu\text{C} = 20 \times 10^{-6} \text{ C}$
 $10 \text{ cm} = 0.1 \text{ m}$

$$\checkmark d = 10\text{cm} = 0.1\text{m}$$

$$\checkmark \Delta V = 0.5\text{KV} = 0.5 \times 10^3\text{V}$$

To find: $E, F = ?$ $V = Ed$

$$E = \frac{\Delta V}{d} = \frac{0.5 \times 10^3\text{V}}{0.1\text{m}}$$

$$E = 5000\text{Vm}^{-1}(\text{NC}^{-1})$$

$E = 5\text{KNc}^{-1}$ ✓

$$F = qE$$

$$F = 20 \times 10^{-6}\text{C} \times 5000\text{NC}^{-1}$$

$$F = 0.1\text{N} = 0.1 \times 10^3 \times (10^{-3})\text{N}$$

$F = 100\text{mN}$

✓ → (12.4)

Q. No 9 (a) What is meant by radioactivity? Compare the properties and behaviour of three types of radiations. α, β, γ → 11.2 Example (5)

(b) A bar 1.0 m in length and located along x-axis moves with a speed of 0.75 c with respect

Q. No 9 (a) What is meant by radioactivity? Compare the properties and behaviour of three types of radiations. α, β, γ $\rightarrow$ 11.2 Example (5)

(b) A bar 1.0 m in length and located along x-axis moves with a speed of 0.75 c with respect to a stationary observer. What is the length of the bar as measured by the stationary observer? (3)